

Prague University of Economics and Business
Faculty of Informatics and Statistics



**Volume, Specificity, and Boilerplate:
A Decomposition of 10-K Risk-Factor
Language and Post-Filing Stock
Return Volatility**

Bachelor Thesis

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Author: Šimon Slanský

Supervisor: prof. RNDr. Ing. Michal Černý, Ph.D.

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Abstract

The Item 1A risk-factor section of the U.S. 10-K is the subject of two parallel literatures whose predictions point in opposite directions. One, represented by **campbell2014**<empty citation>, links a larger *volume* of risk discussion to higher subsequent volatility. The other, represented by **hope2016**<empty citation>, links a higher *specificity* of risk discussion to lower volatility. This thesis takes the question through three layers. It first separates the volume measurement (the log word count of Item 1A) from the specificity measurement (the Loughran–McDonald risk-word density) and enters them jointly in a panel regression of post-filing log-volatility. It then decomposes the composite LM risk-density into its two sub-lists—uncertainty wording and litigious wording—each normalised by Item 1A length, and enters them separately. All specifications use two-digit SIC industry and fiscal-year fixed effects and firm-clustered standard errors. On an unbalanced panel of 535 large U.S. firms over fiscal years 2010–2024 (5,691 firm-years at the 30-day horizon), longer Item 1A sections predict higher post-filing volatility ($\hat{\beta} = +0.130$, $t = +7.70$), while denser Item 1A sections, holding length constant, predict *lower* post-filing volatility ($\hat{\beta} = -3.88$, $t = -2.74$). The LM decomposition reveals that the entire negative effect is carried by the LM-Litigious sub-list ($\hat{\beta}^{\text{lit}} = -6.16$, $t = -3.35$), while the LM-Uncertainty sub-list is statistically indistinguishable from zero. The pattern survives at every horizon from 5 to 365 days and on both COVID-era and pre-COVID sub-samples; we read it in light of the literature on standardised cautionary boilerplate (**nelson2007**; **cazier2021**; **beatty2019**).

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, risk factors, Item 1A, Loughran–McDonald, LM-Litigious word list, boilerplate disclosure, panel data, corporate disclosures

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List of Abbreviations

CIK Central Index Key

DV Dependent Variable

EDGAR Electronic Data Gathering,
Analysis, and Retrieval

EPS Earnings Per Share

FE Fixed Effects

FY Fiscal Year

FYE Fiscal Year End

GLM Generalized Linear Model

L&M Loughran and McDonald

NLP Natural Language Processing

OCF Operating Cash Flow

OLS Ordinary Least Squares

ROA Return on Assets

SEC Securities and Exchange Commission

SIC Standard Industrial Classification

XBRL eXtensible Business Reporting
Language

Introduction

The Item 1A risk-factor section of the U.S. 10-K, mandatory since 2005, is the longest and most carefully written piece of risk-disclosure language in U.S. corporate reporting. Two parallel literatures argue that this section conveys economically meaningful information about firm-level risk, but they reach opposite predictions about its sign. **campbell2014**<empty citation> show that *larger* risk-factor sections, both in raw word count and in the count of risk-related words, are associated with higher subsequent return volatility, market beta, and bid-ask spreads: when managers write more about risk, they are signalling more risk. **hope2016**<empty citation> show that, for a given length, risk-factor sections that contain a higher proportion of *specific* content—names, numbers, geographies—are associated with stronger analyst-forecast revisions and lower implied cost of capital: more concrete risk language pins down exposures and reduces uncertainty.

Much of the empirical literature operationalises both predictions through a single statistic, the dictionary-based risk-word density of Item 1A: the share of the section’s words that match the **loughran2011**<empty citation> uncertainty and litigious word lists. This ratio, however, has section length in its denominator. The volume channel (numerator) and the specificity channel (denominator, inverted) are therefore mixed into one number that has been hard to sign cleanly in the data.

This thesis makes a small methodological move with substantive payoff: it decomposes the dictionary risk-density into its two components—the natural log of the Item 1A word count, and the risk-word density itself—and enters them jointly in a panel regression of post-filing log-volatility. With both regressors in the specification, length captures the volume channel net of any change in boilerplate share, and density captures the specificity channel net of any change in volume. Both predictions can therefore be tested in the same equation, and the data can adjudicate.

On an unbalanced panel of 535 large U.S. non-financial firms over fiscal years 2010–2024 the joint specification at the 30-day horizon delivers exactly the two predictions: longer Item 1A sections predict higher post-filing volatility (the volume effect, H1), while denser Item 1A sections, holding length constant, predict lower volatility (the specificity effect, H2). Both signs and significance survive at every post-filing horizon between 5 and 365 days and across pre-COVID and COVID-era sub-samples; the joint specification also reveals that the dictionary-only risk-density estimate of the specificity channel is roughly twice as large as the true effect, because the omitted volume channel loads onto it through the negative cross-sectional correlation between length and density.

A third contribution looks inside the LM risk lexicon itself. Decomposing the composite RiskDensity into its two LM sub-components—uncertainty-list and litigious-list densities, each normalised by Item 1A length—reveals that the entire negative specificity effect is carried by the LM-Litigious sub-list (H3): conditional on length, LitDensity has a coeffi-

cient of -6.16 ($t = -3.35$) on 30-day post-filing log-volatility, while `UncDensity` is small and statistically indistinguishable from zero. The pattern survives at every horizon and on both COVID sub-samples. The interpretation aligns with the **nelson2007**; **cazier2021**; **beatty2019**<empty citation> reading of litigious-list language as standardised cautionary boilerplate produced under the safe-harbour regime of the 1995 Private Securities Litigation Reform Act and characteristic of mature, legally supervised firm-level disclosure environments.

The remainder of the thesis is organised as follows. Chapter 1 reviews the volume, specificity, and boilerplate literatures and motivates the three hypotheses. Chapter 2 describes the SEC EDGAR data, the textual extraction pipeline, and the construction of the financial controls and the post-filing volatility windows. Chapter 3 formalises the joint specification and the LM-list decomposition. Chapter 4 reports the headline tests of H1, H2, and H3. Chapter 5 reports the multi-horizon evidence, pre/post-COVID splits, a within-firm fixed-effects specification, and a transparent account of seven alternative H3 candidates that were tested and rejected. The Conclusion summarises the contribution and discusses limitations and suggested extensions.

1. Literature Review

This chapter reviews the literature that informs the three questions addressed by the thesis. Section 1.1 surveys the broader programme of textual analysis in accounting and finance and explains why mandatory 10-K disclosures are a natural laboratory for this research. Section 1.2 reviews the strand that links the *volume* of risk-factor disclosure to firm risk, and Section 1.3 reviews the strand that links the *specificity* of risk-factor language to firm risk; together these two literatures motivate H1 and H2. Section 1.4 turns to the internal composition of the Loughran–McDonald (LM) risk lexicon and reviews the relatively orphaned literature on boilerplate cautionary and litigious language, motivating H3. Section 1.5 summarises the financial determinants of volatility used as controls. Section 1.6 states the three hypotheses tested in the rest of the thesis.

1.1 Textual analysis of corporate filings

A now-extensive literature uses computational text analysis to extract information from corporate disclosures. The methodological turning point was **loughran2011**<empty citation>, who showed that the off-the-shelf Harvard General Inquirer dictionary is poorly suited to financial text—roughly three quarters of the words it labels negative are not negative in a financial context—and constructed six word lists (positive, negative, uncertainty, litigious, strong-modal, weak-modal) calibrated on 10-Ks. Their dictionary has become the de-facto standard in accounting research and underpins almost every dictionary-based study cited below. The same authors document that more pessimistic 10-K tone predicts higher one-day filing-period returns, higher trading volume, and higher post-filing volatility, establishing tone as a quantifiable predictor of market outcomes.

Subsequent work expanded the toolkit beyond tone. **li2008**<empty citation> shows that less readable 10-Ks are associated with lower current earnings and weaker earnings persistence. **li2010**<empty citation> estimates a Naive Bayes classifier on forward-looking statements and finds that those with a more positive tone predict higher future earnings. **brown2011**<empty citation> measure the year-on-year textual modification of the MD&A section and document that firms making larger MD&A changes experience larger absolute returns and trading volume around the filing. **dyer2017**<empty citation> apply Latent Dirichlet Allocation to the universe of 10-Ks since 1996 and document a steady increase in disclosure length, much of which is concentrated in boilerplate sections.

Two features of 10-K filings make them an attractive laboratory. First, they are mandatory and standardised: every U.S. public firm files an annual report on essentially the same template, which removes the self-selection bias that complicates voluntary-disclosure studies. Second, the legal regime gives the language teeth: false or misleading statements expose management to private litigation under Rule 10b-5 and to enforcement action by the SEC,

so the words in a 10-K are not cheap talk. Both features apply with particular force to the Item 1A risk-factor section, which is the focus of this thesis.

1.2 The volume channel: more risk language, more uncertainty

The closest paper to the present thesis is **campbell2014**<empty citation>. They examine the Item 1A risk-factor section that the SEC made mandatory in 2005 and argue that managers have private information about firm-specific risks that they reveal in this section. Empirically they show that the *quantity* of risk-factor disclosure—measured both as raw word count and as the count of risk-related words—is positively related to subsequent stock-return volatility, market beta, and bid–ask spreads. Their interpretation is essentially Bayesian: when management chooses to write more about risks, it is signalling that there are more risks, and investors update their assessment of firm-level uncertainty accordingly.

kravet2013<empty citation> reach a similar conclusion for the universe of risk-related disclosures across the 10-K (not only Item 1A): firms with larger increases in the volume of textual risk disclosures experience larger increases in market-assessed risk, measured both as return volatility and as the volume–volatility relation around the filing. **kothari2009**<empty citation>, more broadly, document that the quantity and tone of disclosures by management, analysts, and the business press materially affect cost of capital, return volatility, and forecast dispersion.

The takeaway from this literature is a robust positive association between disclosure *volume* (especially of Item 1A) and subsequent firm-level uncertainty. This motivates Hypothesis H1 of the thesis: holding industry, year, and firm fundamentals fixed, longer risk-factor sections should predict higher post-filing volatility.

1.3 The specificity channel: more concrete language, less uncertainty

A second strand of the literature emphasises that not all words are equal. **hope2016**<empty citation> extract specific risk factors from 10-Ks using a named-entity-recognition pipeline and show that risk-factor disclosures with a higher proportion of *specific* content—names, numbers, geographies—are associated with stronger analyst-forecast revisions, lower forecast dispersion, and lower implied cost of capital. The intuition is that boilerplate risk-factor language conveys little information about the firm’s actual exposures, while specific risk-factor language pins those exposures down and reduces investor uncertainty about them. The two channels can therefore work in opposite directions: more risk language can indicate more risk (the volume channel) but more *informative* risk language can also dispel uncertainty (the specificity channel).

In a panel of large U.S. firms there is no clean named-entity counterpart to **hope2016<empty citation>**'s manual annotation, but the tension can be operationalised in dictionary terms. The Loughran–McDonald uncertainty and litigious word lists capture *generic* risk vocabulary—“may”, “could”, “risks”, “litigation”, “regulatory”—that is widely shared across firms and years. A risk-factor section that is dominated by such generic language has a high *risk-word density*: a large share of its words come from the LM risk lists. A risk-factor section that discusses concrete operational risks (specific suppliers, named regulators, particular products) typically has a lower density of generic risk words, even if its overall length is similar, because the firm-specific nouns and numbers dilute the share of dictionary matches.

Read against **hope2016<empty citation>** this gives a directional prediction. *Holding length constant*, a higher density of generic risk words signals a more boilerplate, less specific section, and should therefore be associated with *lower* post-filing volatility. This is the empirical content of Hypothesis H2.

The two predictions interact in an important way. The dictionary risk-density used in much of the prior literature (**loughran2011**; **kravet2013**) is the *ratio* (risk words)/(total words) inside Item 1A. Because the denominator is itself a measure of section length, this ratio mixes the volume channel (numerator) and the specificity channel (denominator inverted) into one statistic. Sorting them out requires entering both regressors jointly. The thesis therefore tests H1 and H2 in a single specification with $\ln(\text{Words}^{1A})$ and risk-word density as separate explanatory variables.

1.4 Inside the risk lexicon: uncertainty vs. litigious words

The RiskDensity measure used to operationalise H2 collapses two distinct LM word lists into a single share. Following **loughran2011<empty citation>**, the uncertainty list contains words such as “approximate”, “believe”, “contingency”, “depend”, “may”, “might”, “possible”, and “uncertain”, which signal imprecise probability assessments by management. The litigious list contains words such as “allegation”, “claimant”, “defendant”, “indictment”, “litigation”, “plaintiff”, “settlement”, and “subpoena”, which are characteristic of legal-process language. **loughran2015<empty citation>** stress that these are conceptually distinct constructs: uncertainty terms describe *what management does not know about its own business*, whereas litigious terms describe *the legal apparatus that surrounds the business*.

The two sub-lists also have a different production technology in the Item 1A context. Uncertainty language is a natural by-product of candid forward-looking discussion: a firm describing a genuinely open-ended risk uses hedging vocabulary because the underlying uncertainty is real. Litigious language, by contrast, is dominated by standardised cautionary boilerplate that legal counsel inserts in every 10-K of every firm in their portfolio. **nelson2007<empty citation>** show that legal advisers use “meaningful cautionary language” as a safe-harbour shield under the Private Securities Litigation Reform Act of 1995: dense litigious wording is itself an insurance policy against private securities litigation.

cazier2021<empty citation> document that this kind of risk-factor boilerplate is overwhelmingly recycled across years, and that judicial and regulatory assessments treat lengthy boilerplate as an adequate safe-harbour shield even when its concrete information content is low. **beatty2019**<empty citation> reach a similar conclusion for the post-financial-crisis era: market reactions to risk-factor disclosures have weakened over time, consistent with investors discounting the boilerplate component of the section.

Despite this conceptual separation, no panel study to our knowledge has estimated the LM-Litigious density of Item 1A as a stand-alone regressor for post-filing return volatility. **loughran2011**<empty citation> use the four LM lists individually for filing-period returns and trading volume but not for post-filing realised volatility; **kravet2013**<empty citation> aggregate uncertainty- and litigious-related risk language; **campbell2014**<empty citation> hand-code their own risk categories that do not map one-to-one onto the LM dictionary; and **lyle2023**<empty citation> focus on year-on-year changes rather than on the sub-list decomposition. The LM-Litigious sub-list is therefore an orphan in the volatility literature, even though both the production-side argument of **nelson2007**; **cazier2021**<empty citation> and the demand-side argument of **beatty2019**<empty citation> suggest that it should carry a different signal from generic uncertainty wording.

Decomposing the composite RiskDensity into its UncDensity and LitDensity sub-components has a clear directional prediction in light of these papers. Uncertainty wording is a discretionary managerial choice that scales with the genuine ambiguity of the business; conditional on length, it is plausible either to be small in magnitude or to retain the negative sign of the composite RiskDensity (Section 1.3). Litigious wording is closer to a permanent disclosure-style choice—a function of the legal advisers a firm retains and of the maturity of its compliance function—and the **nelson2007**; **cazier2021**<empty citation> literature suggests that firms whose Item 1A is heavier in litigious boilerplate are also firms that operate in more legally-supervised, more institutionalised environments, which is empirically associated with lower realised volatility. The prediction is that, conditional on Item 1A length, LM-Litigious density is the active sub-component of the composite specificity effect, while LM-Uncertainty density either is small in absolute value or absorbs the residual.

1.5 Financial determinants of volatility

The financial controls used in the regressions are standard. Lagged volatility is included to absorb the well-documented persistence of firm-level return variance (**chi2017**). Firm size proxies for diversification and informational depth: larger firms have lower return volatility (**campbell2014**). Leverage captures the financial-distress component of risk; return on assets summarises operating performance, with profitable firms exhibiting lower subsequent volatility (**francis2005**); and asset growth captures the operational-uncertainty channel that follows aggressive expansion. The fixed-effect structure—two-digit SIC industry plus fiscal year—further absorbs persistent industry-level differences in risk and time trends such as the

2008 and 2020 market shocks.

The clustering of standard errors at the firm level follows **petersen2009**<empty citation>; this is essential for inference because the dependent variable is itself a volatility estimate computed on overlapping post-filing windows.

1.6 Hypotheses

The literature reviewed above supports three testable predictions that the thesis evaluates jointly on a panel of 535 large U.S. firms over 2010–2024.

H1 (Volume). Holding firm fundamentals, industry, and year fixed, the natural log of the Item 1A word count is positively associated with post-filing return volatility. This is the **campbell2014**<empty citation> prediction translated to the joint specification.

H2 (Specificity). Holding length and the same controls fixed, the LM risk-word density of Item 1A is *negatively* associated with post-filing return volatility. This is the **hope2016**<empty citation> prediction translated into dictionary terms: once the volume channel is controlled for, a higher share of generic risk vocabulary signals a less specific, more boilerplate section that conveys less new information.

H3 (Boilerplate decomposition). Conditional on Item 1A length and the same controls, the LM-Litigious density of Item 1A carries the entire negative effect attributed to the composite RiskDensity in H2: $\beta^{\text{lit}} < 0$ at the 5% level, while the LM-Uncertainty density is small in absolute value and not statistically distinguishable from zero. The prediction follows from the **nelson2007**<empty citation> and **cazier2021**<empty citation> reading of litigious wording as cautionary boilerplate that is recycled across filings and that proxies for an institutionalised, legally-supervised firm-level disclosure environment.

The three hypotheses are deliberately disciplined by prior work. H1 and H2 settle, in a single specification, an apparent tension between two existing channels (volume vs. specificity) of the risk-factor literature; H3 looks inside the LM risk lexicon to identify which of its two sub-lists is doing the work in H2.

2. Data

2.1 Selecting the Sample

We start with the sample of 1,000 largest U.S. public companies by market capitalisation, taken from the SEC EDGAR company-tickers file, and apply three filters. First, we keep only entities the SEC classifies as “operating,” which removes exchange-traded funds, closed-end funds, blank-cheque companies, and shell entities. Second, we exclude firms in SIC codes 6000–6999 (finance, insurance, and real estate), following common sample-selection practice in prior accounting research (sloan1996; francis2005). Third, we require at least five years of readable XBRL data from 10-K filings and 90% valid stock-return coverage, so that there is enough within-firm variation for fixed-effects estimation.

Sample size. We deliberately focus on large, liquid firms rather than the full EDGAR universe for several reasons. First, our regressions use firm fixed effects with standard errors clustered by firm. As petersen2009<empty citation> shows, the reliability of cluster-robust standard errors depends on the number of clusters—here 539—not on total observations, and 539 clusters are well above the threshold for reliable inference. Also, the textual-analysis part of the thesis requires downloading, parsing, and scoring every 10-K in the sample. Expanding to several thousand firms would greatly increase computation time. Lastly, smaller firms tend to have shorter XBRL histories (mandatory electronic filing was phased in by firm size), more tagging errors, and thinner trading, all of which add noise to financial ratios and return-based variables.

The sample size is in line with prior work combining textual and financial analysis of annual filings. campbell2014<empty citation> study about 9,000 firm-years of risk-factor disclosures from large U.S. firms; loughran2011<empty citation> use a broader set of roughly 50,000 firm-years, but their goal—validating a sentiment dictionary—calls for maximal cross-sectional coverage, whereas the joint volume-specificity test estimated here exploits within-industry-year variation across ~ 535 firms.

2.2 Choosing the Filing Frequency — why annual 10-Ks?

Before moving to variable construction, it is worth explaining why this study uses annual 10-K filings instead of quarterly 10-Q reports. Quarterly data would roughly triple the sample, but two practical problems—shown in Tables 2.1–2.2—ruled out that approach.

Financial data gaps in Q4. 10-Q filings cover only Q1–Q3, so any quarterly panel must reconstruct Q4 as the difference between the annual figure and the sum of the three preceding

quarters. Table 2.1 shows that this residual approach is not usable for operating cash flow: the three quarterly facts coexist in only 7% of firm-years, because most firms report cumulative year-to-date cash flows rather than standalone quarterly amounts. Operating cash flow is the primary input into our OCF-to-assets ratio, so a quarterly panel would lose the cash-flow-based robustness check at the source. Even for net income and operating income, where coverage exceeds 80%, the implied Q4 disagrees with the explicit Q4 by more than one percent of the annual total in about 10–14% of comparable firm-years.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: “Q1 – Q3” indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. “Explicit Q4” indicates the share with a standalone fourth-quarter fact (period ≈ 90 days ending at fiscal year-end). “Comparable” is the count of firm-years for which both an explicit Q4 fact and the implied residual are available. “Match” reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Missing risk-factor text in 10-Q. SEC Regulation S-K only requires firms to update risk factors in quarterly filings when “material changes” have occurred since the last 10-K. In practice, only about half the 10-Q filings in our pilot sample (15 firms, 90 filings) contained a legitimate risk-factor section. The rest had only a reference back to the annual report (Table 2.2). This held across sectors—technology, healthcare, consumer goods, energy—with few exceptions. Constructing the year-on-year text variables that the headline specifications require is therefore not feasible at a quarterly frequency.

Chosen frequency. For these reasons—and following prior textual-analysis work based on 10-K filings (loughran2011; li2008; blankespoor2020)—we use an annual panel. The XBRL coverage comparison (Table A.1) and a consolidated summary (Table A.2) are in Appendix A.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
<i>MD&A (Management’s Discussion and Analysis)</i>		
Section found (%)	93	100
Median word count	20,494	5,806
<i>Risk Factors</i>		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: “Found” requires >200 words extracted by regex-based section parser. “Substantive” for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., “no material changes from those disclosed in our Annual Report”). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

2.3 XBRL Tag Consistency Dilemma

A practical problem with the XBRL Company Facts API is that a single accounting concept can appear under different US-GAAP taxonomy tags. For example, operating cash flow may be reported as either `NetCashProvidedByUsedInOperatingActivities` (total, including discontinued operations) or `...ContinuingOperations` (continuing operations only). Net income may be tagged as `NetIncomeLoss`, `ProfitLoss`, or `IncomeLossFromContinuingOperations`. Stockholders’ equity can include or exclude the non-controlling interest. These tags overlap a lot but are not identical: mixing them all within a single firm’s time series introduces noise, while picking one tag globally would lose firms that consistently report under the other.

We handle this problem with a two-part *tag-harmonisation* process: (i) *equivalence groups* for tags that are economically interchangeable, and (ii) *per-firm tag locking* for tags that are not.

Equivalence groups. When two tags differ only in taxonomy version—not in their economic meaning—we let both enter a firm’s time series. Specifically, the two operating-cash-flow tags and the pair `NetIncomeLoss/ProfitLoss` are treated as interchangeable. The third net-income variant, `IncomeLossFromContinuingOperations`, is *not* included because it deliberately excludes discontinued operations and can differ materially.

To verify this classification, we queried the SEC EDGAR Company Facts API for all 550 sample firms and identified 6,113 firm-year periods in which both tags of a pair were reported simultaneously. The two overlap samples support treating the pairs as interchangeable:

- **Operating cash flow** (1,352 overlaps): 53.5% are numerically identical and the median difference scaled by total assets is practically zero. The 90th-percentile bias is only 1.5 pp.
- **Net income** (4,761 overlaps): 25.5% are numerically identical and the median ROA bias is 0.033 pp. 92.5% of observations lie below 1.0 pp.

The few cases with meaningful differences are already handled by our priority rule, which prefers the parent-company tag whenever both are reported.

Per-firm tag locking. For metrics where the alternative tags *are* economically distinct (stockholders' equity with vs. without NCI; `IncomeLossFromContinuingOperations` vs. `NetIncomeLoss`), we apply strict per-firm tag locking: we keep the tag each firm uses in the most fiscal years and set values from other tags to missing. Even if a firm switches its tag mid-sample, the locked series stays internally consistent at the cost of some discarded observations. This within-firm consistency is what our fixed-effects regressions require.

Combining equivalence groups and per-firm locking increases the complete-case rate for the five regression controls (*roa*, *leverage*, *asset_growth*, *current_ratio*, *log_total_assets*) from 78.9% to 92.7% of firm-year observations. This recovers 988 operating-cash-flow values and 292 net-income values that would otherwise be lost.

Residual limitation. The two-part process does not eliminate every source of cross-tag noise. About 4% of firms end up with a continuing-operations OCF tag paired with a total net-income tag. Ratios that combine the two—such as the auxiliary OCF-to-assets variable used in robustness—carry a small inconsistency. We accept this trade-off in exchange for better coverage. We winsorise these ratios at the 1st and 99th percentiles to clip the resulting extreme values, and note that the same kind of cross-tag mismatch is reported in comparable large-scale XBRL studies ([chychyla2015](#)).

Per-firm tag choices and the overlap diagnostic are reported in Appendix A (Table A.4).

2.4 How do we extract text from 10-K filings?

This study heavily depends on the textual sections of 10-K reports Item 1A *Risk Factors* and Item 7 *Management's Discussion and Analysis of Financial Condition and Results of Operations* (MD&A). This section describes how we find and extract those sections from the raw HTML filings on SEC EDGAR.

2.4.1 Source of textual data

All filings come directly from the SEC EDGAR via the Submissions API, which links each company’s Central Index Key (CIK) to the primary documents of its 10-K filings. From the panel of about 6,700 firm-year observations, we drop 10-K/A amendments. These typically consist of a cover page and a restated exhibit and do not contain the full narrative sections of the original 10-K making them unusable for further analysis. Financial data for amended fiscal years are kept, since the XBRL pipeline already picks up the amended figures when available (Section 2.3). Our extraction thus targets only original 10-K filings.

2.4.2 Section Identification

Unlike structured XBRL data, the textual parts of 10-K filings are hidden inside HTML with no standardised tagging. A raw filing can span across several thousands of lines, with section headers defined in many different ways: bold text, explicit `<font>` tags, inline uppercase text, or table-based layouts. Additionally, these headers are often mixed in with a table of contents (TOC), running headers, and inline cross-references (e.g. “see Item 1A”). These features make simple keyword search on plain text basically impossible, because it confuses TOC entries, page headers, and inline mentions with actual section boundaries.

To achieve the best possible results, we use an HTML extraction procedure with four stages:

1. **Header detection.** We walk through the HTML at the block level and test each element against a set of regex patterns for the target sections (e.g. a pattern that matches “*Item 1A*” together with its common variants). A candidate for correct section counts only if it meets at least one formatting rule: bold styling, ALL-CAPS text, or large font size (≥ 12 pt). This effectively filters out inline cross-references, which sit in normal body-text paragraphs with no special formatting, without discarding any valid headers. Elements that are themselves hyperlinks are also rejected, as these are almost always table-of-contents entries pointing to the real section.
2. **Position marking.** Each confirmed header is tagged in place before the document is flattened to plain text, so we keep track of where every header sits in the final text.
3. **Candidate selection.** For each target section, we keep only headers whose surrounding text contains an expected subtitle keyword (“risk” for Item 1A; “management” or “discussion” for Item 7) and discard any header having many other Item headers near it (within 300 characters in either direction), the typical signature of a table of contents. When several candidates remain, we pick the one followed by the longest stretch of text before the next section header.
4. **Section boundary.** The section text runs from the selected header to the first occurrence of the next expected Item header (Item 1B, 1C, or 2 for Item 1A; Item 7A or 8 for Item 7). Sections shorter than 200 words are treated as stubs and trigger a fallback procedure described in the next subsection.

2.4.3 Exhibit Fallback

A common practice in 10-K filings—especially older ones—is to replace an Item with a short stub such as “This information is incorporated by reference from Exhibit 13.” When the extracted section has fewer than 200 words and the text matches an incorporation-by-reference pattern, the parser finds the referenced exhibit number (defaulting to Exhibit 13, the annual report to shareholders) and downloads the matching HTML document from the filing index. The section identification procedure described in the previous subsection is then rerun on the exhibit. Because exhibits often skip the formal “Item X” prefix, the parser also matches patterns with titles only (e.g. “Risk Factors”, “Management’s Discussion and Analysis”) as a backup detection layer for both the main document and exhibits.

2.4.4 Post-Processing

After converting the HTML to plain text, several layout artefacts from the original filing remain. These could potentially harm later text analysis. Before computing any textual features, we perform 3 cleaning steps that remove three types of noise:

1. **Running page headers.** Many filings embed “*<page number> Table of Contents*” strings at page boundaries. A simple regular expression strips any line matching a one- to three-digit number followed by “Table of Contents.”
2. **Standalone page numbers.** After removing the full header pattern, some filings still contain page numbers on empty lines. These are also removed by deleting any line consisting solely of a one- to three-digit integer.
3. **Non-breaking spaces.** Non-breaking-space characters introduced by the original HTML markup are normalised to standard spaces, so that downstream word-count and dictionary-based measures are not affected by these artefacts.

The cleaning rules are intentionally conservative: they target only patterns that cannot realistically be part of meaningful text and are applied after section boundaries have already been set, so they do not interfere with header detection.

2.4.5 Coverage and Limitations

Because the quality of the extraction pipeline is crucial for downstream analysis, we validated it on two samples: the 50 most recent filings in the panel (fiscal year 2024) and, more importantly, a stratified random sample of 100 filings covering all fiscal years 2010–2024 (20 per era). On the recent sample, the parser successfully extracted both sections for all 50 filings (100%). On the cross-era sample, 97 of 100 filings yielded both sections. The three failures break down as follows:

- Two filings whose annual report exhibit was filed as a PDF rather than HTML, making

it inaccessible to an HTML parser;

- One filing whose 10-K used neither bold, large-font, nor uppercase cues for its section headers, leaving no formatting signal for the parser to use.

Neither failure mode is related to firm characteristics in any systematic way. On the full panel, we expect a coverage rate of roughly 97%, with missing values treated as missing in the textual variable construction.

LLM-assisted quality audit. Although the parser extracted 97/100 filings successfully, coverage alone says nothing about content quality. To check whether the extracted sections are actually the right ones, we ran an independent validation using a large language model (Google Gemini 2.5 Flash, accessed through its OpenAI-compatible API, temperature = 0 for deterministic results). [gilardi2023](#) show that LLM-based annotation matches or surpasses trained crowd-workers on text-classification tasks, which supports using a language model as an independent extraction validator. A stratified random sample of 100 filings was drawn—20 filings from each of five time periods (2010–2012, 2013–2015, 2016–2018, 2019–2021, 2022–2024). Both Item 1A and Item 7 were evaluated for every filing, giving 200 section-level evaluations in total. This design is in line with what [krippendorff2019](#) recommends as reliable for content-analysis studies, and large enough that pass-rate estimates carry a margin of error of about ± 3 percentage points. The 100 filings also cover roughly 19% of the 539-firm panel and span the full history of SEC filing formats.

For each filing, both extracted sections were sent to the model in a single prompt along with a structured evaluation rubric. The model evaluated each extracted section on five criteria—correct section identity, start-boundary accuracy, end-boundary accuracy, content quality (no residual HTML tags, page headers, or distorted data), and completeness (full narrative, not a stub or truncated fragment)—using a three-level grading scale: *Pass* (criterion fully met), *Minor* (small issue that would not significantly affect NLP analysis), and *Fail* (issue that would significantly affect NLP analysis). Additionally, we report two aggregate metrics: a strict rate (only *Pass*) and a lenient rate (*Pass* + *Minor*), each with Wilson 95% confidence intervals ([agresti1998](#)).

Table 2.3: LLM-assisted extraction validation results ($n = 200$ sections, 100 per section type).

Criterion	Strict	%	Lenient	%	95% CI (lenient)
Correct section	192/200	96.0	192/200	96.0	[92.3–98.0%]
Start boundary	191/200	95.5	193/200	96.5	[93.0–98.3%]
End boundary	192/200	96.0	193/200	96.5	[93.0–98.3%]
Content quality	191/200	95.5	195/200	97.5	[94.3–98.9%]
Completeness	189/200	94.5	191/200	95.5	[91.7–97.6%]
Overall (all five)	179/200	89.5	187/200	93.5	[89.2–96.2%]

Table 2.3 shows the results. On every individual criterion, the lenient pass rate (*Pass* + *Minor*)

is above 95%, confirming that the parser finds the right section with accurate boundaries and produces clean, complete text. Even under the strict metric, the rates stay above 94% on each criterion. The overall lenient rate across all five criteria at once is 93.5% (187/200; 95 % CI: 89.2–96.2 %).

Of the 13 sections with at least one criteria resulting in Fail, five were complete extraction failures (empty output) caused by non-standard HTML layouts. Four had wrong section boundaries (content from an adjacent Item was included). Finally, the remaining four were flagged only for low word counts suggesting possible truncation, which we do not permit to enter the main analysis. Item 7 extractions do slightly better than Item 1A (lenient pass rates of 96% vs. 91%), most likely because Item 7 has an advantage in cleaner end boundary (Item 7A or Item 8), while Item 1A sometimes wrongly picks up text from the preceding Item 1.

Failures clearly cluster in older filings: 5 of the 40 sections from 2010–2012 were flagged. On the other hand, only 1 of 40 come from 2022–2024. This shows expected gradual standardisation of SEC filing HTML over the sample period.

Robustness of the LLM-based validation. Several choices in this validation procedure support the reliability of this approach. First, the validation is built around clearly defined, verifiable criteria—correct section identity, boundary accuracy, absence of HTML artefacts, and word-count thresholds—that leave little room for subjective interpretation and ambiguity. Second, each judgment covers a single document with a deterministic prompt. The model does not need to compare across filings or aggregate statistics. Third, manually checking the model’s Fail verdicts and reasons confirmed agreement in every case examined.

Relation to prior approaches. Our extraction strategy is inspired by [dyer2017](#)<empty citation>, who document substantial evolution in 10-K disclosure—length, boilerplate, and structure—over time and the challenges this creates for large-sample studies. Compared with plain-text methods used in earlier work ([li2008](#); [loughran2011](#); [campbell2014](#)), our procedure avoids false positives from TOC entries and inline cross-references. Compared with XBRL-tagged textual blocks (e.g. `us-gaap:RiskFactorsTextBlock`), which became available with Inline XBRL adoption, our method covers a wider time span: structured textual tags are inconsistently filled before 2018, while the HTML-based approach works across the full 2010–2024 window.

3. Methodology

This chapter explains how the data described in Chapter 2 are used to test the three hypotheses set out in Section 1.6. Section 3.1 gives an overview of the empirical strategy. Section 3.2 defines the regression variables and Section 3.3 formalises the joint specification used for H1 and H2 and the LM-list-decomposition specification used for H3. Section 3.4 describes estimation and inference, and Section 3.5 states the formal decision rules used to evaluate each hypothesis. Section 3.6 previews the robustness checks reported in Chapter 5.

3.1 Empirical strategy

The thesis tests two competing predictions about the same section of the 10-K—Item 1A, the mandatory risk-factor disclosure—in a single regression specification. The volume channel ([campbell2014](#)) predicts that longer risk-factor sections indicate more risk and therefore higher post-filing volatility. The specificity channel ([hope2016](#)) predicts that, holding length constant, a higher share of *generic* risk vocabulary indicates a more boilerplate and less informative section, and is therefore associated with *lower* post-filing volatility. Because the dictionary risk-density commonly used in prior work has section length in its denominator, separating the two channels requires entering log word count and risk-word density jointly.

Once the joint specification is in place, H3 looks inside the composite RiskDensity. The same headline regression is re-estimated with the LM uncertainty- and litigious-list densities entered separately as UncDensity and LitDensity, and the two coefficients are inspected for which sub-list is the active ingredient of the H2 effect. By construction $\text{RiskDensity} = \text{UncDensity} + \text{LitDensity}$, so the decomposition is an algebraic refinement of H2, not a competing specification. The same six post-filing horizons ($h \in \{5, 10, 30, 90, 180, 365\}$ days) are used in the robustness chapter to confirm that the sub-list that is identified as active at the headline horizon retains its sign and significance across all alternative windows.

The dependent variable is the natural log of post-filing annualised volatility. The log transformation has two advantages. First, raw volatility is bounded below by zero and right-skewed, so OLS in levels gives disproportionate weight to the right tail; the log transformation symmetrises the distribution and makes coefficients on continuous regressors directly interpretable as semi-elasticities. Second, the log transformation makes the specification scale-free: a one-unit change in $\ln(\text{Words}^{1A})$ corresponds to a constant percentage change in volatility regardless of the firm’s baseline level of risk.

3.2 Variables

3.2.1 Dependent variable: post-filing log-volatility

The dependent variable is the natural logarithm of the annualised standard deviation of daily log returns computed over a post-filing window of length h that begins two trading days after the 10-K filing date:

$$\ln \sigma_{i,t+1}^{[h]} = \ln \left(\sqrt{252} \cdot \text{sd}(r_{i,d})_{d \in W_{i,t}^{[h]}} \right), \quad (3.1)$$

where $r_{i,d} = \ln(P_{i,d}/P_{i,d-1})$ is the daily log return for firm i and $W_{i,t}^{[h]}$ is the post-filing window of length h calendar days for filing year t . The two-day lag avoids the same-day price reaction to the filing itself; the $\sqrt{252}$ factor rescales the daily standard deviation to an annualised figure, which is the conventional unit in the volatility literature.

For the headline specification we use $h = 30$ days, which is short enough to capture the post-filing reaction documented by [loughran2011](#) but long enough that the volatility estimate is built on a meaningful number of daily returns (≈ 21 trading days). Robustness checks at $h \in \{5, 10, 90, 180, 365\}$ trace the temporal decay of the signal (Section 3.6 and Chapter 5).

3.2.2 Financial controls

The controls are taken directly from the firm’s annual XBRL filings (Section 2.3). Five regressors enter every specification: *lagged log-volatility* $\ln \sigma_{i,t}^{[h]}$ (the same measure as the dependent variable but for the previous filing year’s window of identical length), *firm size* $\ln(\text{Total Assets})$, *leverage* (total liabilities over total assets), *return on assets* (net income over total assets), and *asset growth* (year-on-year change in total assets divided by lagged total assets). All ratios are winsorised at the 1st and 99th percentiles within the regression sample to bound the influence of extreme values without discarding economically meaningful observations.

3.2.3 Text variables: Item 1A length and risk-word density

Two text variables operationalise the two channels of Sections 1.2–1.3.

Item 1A length (H1). The volume channel is captured by the natural logarithm of the cleaned word count of Item 1A:

$$\text{Words}_{i,t}^{1A} = \text{number of tokens in Item 1A of firm } i\text{'s year-}t \text{ 10-K}, \quad (3.2)$$

$\ln(\text{Words}_{i,t}^{1A})$ enters the regression. The cleaning pipeline (Section 2.4.4) strips XBRL tags, footnote markers, and table cell contents before tokenisation, so the count reflects genuine

narrative content rather than markup. Across the sample the median Item 1A contains roughly 8,800 words, with a long right tail (Section 4.1).

Item 1A risk-word density (H2) and its decomposition (H3). The specificity channel is captured by the share of words in Item 1A that match the `loughran2011<empty citation>` uncertainty or litigious word lists:

$$\text{RiskDensity}_{i,t} = \frac{n_{i,t}^{\text{unc}} + n_{i,t}^{\text{lit}}}{\text{Words}_{i,t}^{1A}}, \quad (3.3)$$

where n^{unc} and n^{lit} are the per-section counts of uncertainty- and litigious-list words. Higher values indicate that a larger share of the section consists of generic risk vocabulary; lower values indicate a section in which the narrative is dominated by firm-specific or non-dictionary content—names, numbers, geographies, products—that the `hope2016<empty citation>` reading associates with greater specificity.

For H3 the same composite is decomposed into its two LM sub-lists, each normalised by Item 1A length:

$$\text{UncDensity}_{i,t} = \frac{n_{i,t}^{\text{unc}}}{\text{Words}_{i,t}^{1A}}, \quad (3.4)$$

$$\text{LitDensity}_{i,t} = \frac{n_{i,t}^{\text{lit}}}{\text{Words}_{i,t}^{1A}}, \quad (3.5)$$

with the algebraic identity $\text{RiskDensity} = \text{UncDensity} + \text{LitDensity}$. The two sub-densities operationalise the conceptually distinct uncertainty and litigious constructs of Section 1.4: UncDensity captures hedging vocabulary about what management does not know about its own business, while LitDensity captures the cautionary legal-process boilerplate that legal advisers insert under the safe-harbour regime of `nelson2007<empty citation>`.

3.2.4 Why both variables are needed

The two variables are mechanically related but not redundant. The denominator of RiskDensity is the same total word count that enters H1 in logs. Across the sample the Pearson correlation between $\ln(\text{Words}^{1A})$ and RiskDensity is -0.31 , which is far from collinear but is also not zero. Entering the two variables jointly therefore lets each coefficient be interpreted holding the other fixed: β^{len} measures the volatility response to a longer Item 1A at *constant* risk-word density (the volume channel net of any change in boilerplate share), and β^{dens} measures the volatility response to a denser Item 1A at *constant* length (the specificity channel net of any change in volume). Variance-inflation factors for both variables are below 1.5 in the joint specification, well within conventional thresholds.

This joint specification of length and density is one of the methodological points of the thesis. As discussed in Section 1.3, the dictionary-only risk measure used in much prior work

mixes the two channels into one statistic; entering them jointly recovers two opposite-signed associations that partially cancel when only the ratio is examined.

3.3 Model specifications

Let $\mathbf{X}_{i,t}^{\text{fin}}$ collect the four financial controls (size, leverage, ROA, asset growth) and let $\sigma_{i,t}^{[h]}$ denote the firm's lagged volatility computed on the prior filing year's window of the same length h .

Joint specification (H1 and H2).

$$\ln \sigma_{i,t+1}^{[h]} = \alpha + \phi \ln \sigma_{i,t}^{[h]} + \beta' \mathbf{X}_{i,t}^{\text{fin}} + \beta^{\text{len}} \cdot \ln(\text{Words}_{i,t}^{1A}) + \beta^{\text{dens}} \cdot \text{RiskDensity}_{i,t} + \mu_j + \tau_t + \varepsilon_{i,t}, \quad (3.6)$$

where μ_j is a two-digit SIC industry fixed effect and τ_t is a fiscal-year fixed effect. Equation 3.6 is estimated at the headline horizon $h = 30$ days for the main table (Section 4.2) and at $h \in \{5, 10, 30, 90, 180, 365\}$ for the multi-horizon table 4.5. H1 is supported if $\beta^{\text{len}} > 0$ and significant at the 5% level; H2 is supported if $\beta^{\text{dens}} < 0$ and significant at the 5% level. Both predictions must hold jointly in equation 3.6 for the volume-specificity decomposition to be considered confirmed.

To document the marginal contribution of each text variable, the results chapter also reports two partial specifications that drop β^{dens} (column 2 of the headline table) or β^{len} (column 3) from equation 3.6. The five-column nested table (baseline / +length / +density / +both / +length+decomposition) makes visible the fact that each variable carries meaningful incremental information and that the composite RiskDensity hides a heterogeneous mix of two LM sub-components, only one of which is active.

Decomposition specification (H3). For H3 the composite RiskDensity in equation 3.6 is replaced by its two LM sub-densities entered jointly:

$$\ln \sigma_{i,t+1}^{[h]} = \alpha + \phi \ln \sigma_{i,t}^{[h]} + \beta' \mathbf{X}_{i,t}^{\text{fin}} + \beta^{\text{len}} \cdot \ln(\text{Words}_{i,t}^{1A}) + \beta^{\text{unc}} \cdot \text{UncDensity}_{i,t} + \beta^{\text{lit}} \cdot \text{LitDensity}_{i,t} + \mu_j + \tau_t + \varepsilon_{i,t} \quad (3.7)$$

By the algebraic identity $\text{RiskDensity} = \text{UncDensity} + \text{LitDensity}$, the decomposition (equation 3.7) nests the joint specification (equation 3.6) as the special case $\beta^{\text{unc}} = \beta^{\text{lit}}$. Equation 3.7 is estimated at the headline $h = 30$ days for column (5) of the headline table (Section 4.2) and across all six post-filing horizons in Table 4.5. H3 is supported if $\beta^{\text{lit}} < 0$ at the 5% level with the sign and significance preserved at every horizon, while β^{unc} is small in absolute value and statistically indistinguishable from zero. The Pearson correlation between UncDensity and LitDensity is essentially zero ($r \approx -0.05$ in the regression sample), so the two coefficients are separately identified.

3.4 Estimation and inference

Equation 3.6 is estimated by ordinary least squares with two-digit SIC industry and fiscal-year fixed effects absorbed using the absorbing-regression routine of the `linearmodels` package. Coefficients are therefore identified from *within-industry*, *within-year* variation, which removes the risk that the results are driven by either time trends in volatility (e.g. the 2008 and 2020 crises) or by persistent industry-level differences in risk vocabulary or risk levels.

Standard errors. Standard errors are two-way clustered by firm and fiscal year following `petersen2009` and `cameron2011robust`, which makes inference robust to arbitrary within-firm correlation *and* to arbitrary within-year cross-sectional correlation in the residuals. Three features of the data make the year dimension important alongside the firm dimension. First, the dependent variable is itself a volatility estimate computed on a rolling post-filing window, and consecutive firm-year windows from a single firm overlap. This induces strong serial correlation in $\ln \sigma_{i,t+1}$ that ordinary heteroskedasticity-robust standard errors would understate. Second, omitted firm-specific risk characteristics are likely to persist over time. Third, calendar-year shocks (most visibly the 2020 COVID shock) generate strong cross-sectional correlation in residuals within a fiscal year that one-way firm clustering would miss. Two-way clustering addresses all three concerns simultaneously without imposing a particular correlation structure.

Outliers. The financial ratios used as regressors are winsorised at the 1st and 99th percentiles of the regression sample (Section 3.2.2). The text variables are not winsorised: $\ln(\text{Words}^{1A})$ is naturally bounded because of the log transformation and RiskDensity is a ratio bounded between zero and one in practice (sample maximum ≈ 0.05).

Sample composition. After applying all data filters described in Chapter 2 and dropping firm-years with missing values for any regressor, the estimation sample at the 30-day horizon consists of 5,691 firm-years on 535 unique firms. The 365-day horizon retains 6,172 firm-years on the same set of firms; the larger sample reflects the fact that the longer window admits more observations near the end of the panel period for which a 30-day window does not yet have a complete 30 trading days of post-filing returns.

3.5 Testing the hypotheses

H1 (Volume). Supported if $\hat{\beta}^{\text{len}} > 0$ at the 5% level in equation 3.6 estimated at $h = 30$ days, and the sign survives in the 365-day specification and at every intermediate horizon. The economic magnitude is reported as the percentage change in $\sigma_{i,t+1}^{[30]}$ implied by an interquartile-range movement in $\ln(\text{Words}^{1A})$.

H2 (Specificity). Supported if $\hat{\beta}^{\text{dens}} < 0$ at the 5% level in the same joint specification, with the sign and significance preserved across horizons. The magnitude is reported analogously.

H3 (Boilerplate decomposition). Supported if, in equation 3.7 estimated at $h = 30$ days, $\hat{\beta}^{\text{lit}} < 0$ at the 5% level and $\hat{\beta}^{\text{unc}}$ is statistically indistinguishable from zero. The qualitative pattern (LitDensity strongly negative, UncDensity null) must survive at every horizon $h \in \{5, 10, 30, 90, 180, 365\}$ and on both sub-period samples (Section 3.6).

The three criteria are evaluated jointly. Statistical significance alone is not enough: results that are statistically significant in the headline 30-day specification but reverse sign at neighbouring horizons would be evidence of overfitting rather than of a robust economic channel.

3.6 Planned robustness checks

Chapter 5 reports a battery of checks intended to ensure that the headline results do not depend on a single modelling choice. Briefly, equation 3.7 is re-estimated (i) at five additional post-filing horizons (5, 10, 90, 180, and 365 days) to confirm that the LM-Litigious sub-list is the active sub-component at every window; (ii) separately on the pre-COVID (2010–2019) and COVID-era (2020–2024) sub-periods, to confirm that the result is not driven by either macro regime; (iii) with firm fixed effects in place of industry fixed effects, which absorbs all time-invariant firm-level heterogeneity and isolates within-firm year-to-year variation. Each check is reported in full in the tables of Chapter 5.

4. Results

This chapter reports the empirical results. Section 4.1 summarises the regression sample. Section 4.2 presents the headline test of H1 and H2 at the 30-day horizon and introduces the H3 boilerplate-decomposition column. Section 4.3 reproduces the same specification at the 365-day horizon to confirm that the results are not an artefact of the short-window choice. Section 4.4 runs the H1+H3 specification at every horizon between 5 and 365 days, and Section 4.5 discusses the H3 evidence formally. Section 4.6 translates the estimated coefficients into economic magnitudes.

4.1 Descriptive statistics

Table 4.1 reports summary statistics for the regression sample at the 30-day horizon. The dependent variable is post-filing volatility in logs (equation 3.1); its distribution is approximately symmetric with mean $\ln \sigma_{i,t+1}^{[30]} \approx -1.34$ and standard deviation ≈ 0.59 , and an interquartile range running from -1.74 to -1.06 . In raw units this corresponds to a typical annualised post-filing volatility around 0.26 and a P25–P75 band of roughly 0.18–0.35, in line with the right-skewed distribution typical of equity volatility.

Lagged log-volatility has a near-identical distribution as expected for a persistent series. Firm size is measured as the natural log of total assets in dollars, with a sample mean of 23.18 (roughly \$11.7 billion in assets) and an interquartile range from 22.19 to 24.24 (roughly \$4.4 to \$33.6 billion); this confirms the focus on large, liquid firms motivated in Section 2.1. Leverage centres around 0.62 with moderate spread; ROA averages 0.06 with a long left tail; and asset growth averages 0.11 with a long right tail driven by acquisitive firms.

The two text variables show economically meaningful spread. $\ln(\text{Words}^{1A})$ has a mean of 9.03 and a standard deviation of 0.64, corresponding to a median Item 1A of roughly 8,800 words and an interquartile range from $\approx 5,650$ to $\approx 12,300$ words. RiskDensity has a mean of 0.051 and a standard deviation of 0.006: roughly 5% of the words in a typical Item 1A come from the LM uncertainty or litigious lists. The two LM sub-components used for H3 are unevenly sized: the uncertainty share averages 0.034 ($\approx 68\%$ of RiskDensity) and the litigious share averages 0.016 ($\approx 32\%$). The two sub-shares are essentially uncorrelated ($r \approx -0.05$ in the regression sample), confirming that the decomposition delivers two genuinely separate regressors.

Table 4.2 reports pairwise Pearson correlations. Lagged log-volatility is the strongest single predictor of log-volatility ($r = 0.35$). Larger firms have lower log-volatility ($r \approx -0.19$) and higher profitability also tracks lower log-volatility ($r \approx -0.23$), in line with the well-known size–risk and earnings–quality relationships. The two text variables are correlated at $r = -0.30$: longer Item 1A sections tend to have lower risk-word density, consistent with their

Table 4.1: Summary Statistics

	Mean	SD	P25	Median	P75
$\ln(\sigma_{i,t+1})$	-1.344	0.588	-1.740	-1.410	-1.058
$\ln(\sigma_{i,t})$	-1.353	0.590	-1.749	-1.424	-1.065
$\ln(\text{Total Assets})$	23.182	1.452	22.191	23.206	24.236
Leverage	0.616	0.218	0.476	0.611	0.739
ROA	0.063	0.087	0.027	0.062	0.106
Asset Growth	0.113	0.243	0.001	0.059	0.142
$\ln(\text{Words}^{1A})$	9.026	0.637	8.639	9.079	9.418
RiskDensity	0.051	0.006	0.047	0.051	0.055
UncDensity	0.034	0.005	0.032	0.034	0.037
LitDensity	0.017	0.005	0.013	0.016	0.019
Observations	5,691				
Firms	535				

Note: Summary statistics for the regression sample at the 30-day horizon. $\sigma_{i,t+1}$ is the annualised standard deviation of daily log returns over the 30-day window starting two trading days after the filing date; $\ln(\sigma_{i,t+1})$ is its natural logarithm. Words^{1A} is the cleaned word count of Item 1A; RiskDensity is the share of LM uncertainty- and litigious-list words in Item 1A; UncDensity and LitDensity are the two sub-components separately. Financial ratios are winsorised at the 1st and 99th percentiles.

narrative being padded by firm-specific text rather than by extra generic risk vocabulary. The negative correlation is moderate and leaves both signals separately identifiable in the joint specification.

Table 4.2: Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) $\ln(\sigma_{i,t+1})$	1							
(2) $\ln(\sigma_{i,t})$	0.35***	1						
(3) $\ln(\text{Total Assets})$	-0.19***	-0.20***	1					
(4) Leverage	-0.06***	-0.07***	0.21***	1				
(5) ROA	-0.23***	-0.25***	0.05***	-0.06***	1			
(6) Asset Growth	0.16***	0.09***	-0.12***	-0.12***	-0.04***	1		
(7) $\ln(\text{Words}^{1A})$	0.31***	0.31***	-0.09***	-0.04***	-0.27***	0.17***	1	
(8) RiskDensity	-0.11***	-0.12***	-0.01	-0.01	0.06***	-0.08***	-0.30***	1
(9) UncDensity	-0.09***	-0.08***	-0.18***	-0.06***	0.14***	-0.05***	-0.43***	0.71***
(10) LitDensity	-0.06***	-0.08***	0.18***	0.04***	-0.06***	-0.06***	0.03**	0.66***

Note: Pairwise Pearson correlation coefficients on the 30-day-horizon regression sample. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.2 Volume, specificity, and boilerplate at the 30-day horizon

Table 4.3 reports the headline test of H1, H2, and H3. Column (1) is the financial baseline; Column (2) adds Item 1A length in logs (the volume channel); Column (3) adds the composite risk-word density (the specificity channel); Column (4) is the joint specification of equation 3.6 that nests H1 and H2; Column (5) is the H3 decomposition specification of equation 3.7, which replaces RiskDensity by its two LM sub-components.

The baseline (Column 1). Lagged log-volatility carries the expected positive coefficient (+0.170, $t = +2.12$ under two-way firm-and-year clustering): a one-percent rise in last-year’s volatility maps into about 0.17% extra volatility this year. The relatively small t -statistic on the lagged dependent variable reflects the strong residual year-level comovement in volatility (e.g. the 2020 COVID shock) that year clustering correctly accounts for. Firm size enters negatively (-0.070 , $t = -5.81$): larger firms are less volatile. Profitability is strongly stabilising (ROA coefficient -1.07 , $t = -9.72$); asset growth predicts higher future volatility (+0.246, $t = +4.34$); leverage is statistically indistinguishable from zero, as in much of the volatility literature (campbell2014). The baseline already explains 49% of the in-sample variation in log-volatility.

The volume channel (Columns 2, 4, and 5). Adding $\ln(\text{Words}^{1A})$ alone (Column 2) raises adjusted R^2 to 0.505 and yields a length coefficient of +0.138 ($t = +8.43$). In the joint specification (Column 4), where the composite risk-word density is controlled for, the length coefficient is +0.121 ($t = +7.64$). In the H3-decomposition specification (Column 5), where the composite is replaced by its two LM sub-components, the length coefficient is +0.130 ($t = +7.16$). The pattern is consistent across specifications: longer risk-factor sections are

Table 4.3: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (30 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.1695** (+2.12)	+0.1368* (+1.80)	+0.1552** (+1.96)	+0.1336* (+1.76)	+0.1321* (+1.75)
$\ln(\text{Total Assets})$	-0.0700*** (-5.81)	-0.0671*** (-5.91)	-0.0695*** (-5.99)	-0.0672*** (-5.98)	-0.0645*** (-5.69)
Leverage	-0.0198 (-0.56)	-0.0026 (-0.07)	-0.0249 (-0.72)	-0.0072 (-0.21)	-0.0082 (-0.24)
ROA	-1.0678*** (-9.72)	-0.8982*** (-8.68)	-1.0689*** (-9.80)	-0.9189*** (-8.80)	-0.9250*** (-8.77)
Asset Growth	+0.2460*** (+4.34)	+0.2033*** (+3.80)	+0.2313*** (+4.21)	+0.2011*** (+3.78)	+0.1982*** (+3.72)
$\ln(\text{Words}^{1A})$		+0.1375*** (+8.43)		+0.1212*** (+7.64)	+0.1298*** (+7.16)
RiskDensity			-7.7820*** (-5.40)	-3.8769*** (-2.68)	
UncDensity					-1.2779 (-0.62)
LitDensity					-6.1591*** (-3.38)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,691	5,691	5,691	5,691	5,691
Firms	535	535	535	535	535
Adj. R^2	0.492	0.505	0.497	0.506	0.507

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 30 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its decomposition (H3) into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t -statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

associated with higher post-filing log-volatility at the 30-day horizon. The point estimate of 0.130 from Column (5) corresponds to roughly an 8% increase in post-filing volatility for a one-standard-deviation rise in $\ln(\text{Words}^{1A})$ ($0.64 \times 0.130 = 0.083$, applied to $\ln \sigma$). **H1 is supported.**

The specificity channel (Columns 3 and 4). Adding RiskDensity alone (Column 3) yields a coefficient of -7.78 ($t = -5.40$). In the joint specification, where the volume channel is controlled for, the coefficient remains negative and significant at -3.88 ($t = -2.68$, $p = 0.007$). The fact that the density coefficient halves once length is controlled for is exactly the mechanical pattern anticipated in Section 3.2.4: about half of what looked like a “risk-density effect” in Column (3) was in fact the volume effect operating through the section’s word-count denominator. Once $\ln(\text{Words}^{1A})$ is partialled out, the residual density effect is purely specificity in the **hope2016**<empty citation> sense: a denser, more boilerplate Item 1A is associated with *lower* post-filing volatility. **H2 is supported**, and the joint specification reveals an effect that the dictionary-only ratio used in much prior work would have hidden.

The boilerplate decomposition (Column 5). Replacing the composite RiskDensity by its two LM sub-components reveals that the negative specificity association of Column (4) is carried almost entirely by the LM-Litigious sub-list. LitDensity enters with a coefficient of -6.16 ($t = -3.38$, $p < 0.001$), while UncDensity enters with a small and statistically insignificant coefficient of -1.28 ($t = -0.62$, $p = 0.53$). A formal Wald test of $H_0 : \beta^{\text{unc}} = \beta^{\text{lit}}$ yields $z = +1.91$ ($p = 0.056$), so the two coefficients are distinguishable at the 10% but not the 5% level under the full two-way clustering scheme. The 95% confidence interval on β^{unc} is wide ($[-5.30, +2.74]$): the data are consistent with a modestly negative uncertainty effect, but cannot rule out zero. Adjusted R^2 in Column (5) is essentially identical to that of the composite specification in Column (4) (0.5068 vs. 0.5064), because the litigious sub-list captures the variance that drives the composite. The H2 specificity association is therefore concentrated in the litigious rather than the uncertainty sub-list, consistent with the cautionary, legal-process language that **nelson2007**; **cazier2021**<empty citation> link to mature, legally supervised disclosure environments. **H3 is supported in the cross-section**, with the qualifications discussed in Section 4.5 and the firm-FE caveat of Section 5.3.

Joint significance and incremental fit. The joint test of $H_0 : \beta^{\text{len}} = \beta^{\text{unc}} = \beta^{\text{lit}} = 0$ in Column (5) rejects at the 1% level. Adding the three text variables to the baseline raises adjusted R^2 from 0.492 (Column 1) to 0.507 (Column 5), an increment that is small in absolute terms because the financial baseline already captures the bulk of predictable volatility but is large by the standards of the textual-disclosure literature (**kravet2013**; **campbell2014**). The three text variables are informative both individually and jointly.

4.3 The same channels at the 365-day horizon

Table 4.4 re-estimates the same five nested specifications with the dependent variable replaced by $\ln \sigma_{i,t+1}^{[365]}$, the log of annualised volatility over the full year following the filing. The lagged dependent variable is the analogous one-year window for the prior filing year.

The signs and significance pattern of all three text variables survive at the annual horizon. In the H1+H2 joint specification (Column 4) $\beta^{\text{len}} = +0.056$ ($t = +4.53$) and $\beta^{\text{dens}} = -2.19$ ($t = -3.55$, $p < 0.001$). In the H1+H3 decomposition (Column 5) $\beta^{\text{len}} = +0.061$ ($t = +4.93$), $\beta^{\text{lit}} = -3.39$ ($t = -3.91$, $p < 0.001$), and $\beta^{\text{unc}} = -0.86$ ($t = -1.08$, $p = 0.28$). What changes between the 30-day and 365-day windows is the magnitude: the length coefficient is roughly half of the 30-day estimate; the litigious coefficient is roughly 55% of the 30-day estimate. Both shrink toward zero as the volatility window lengthens, exactly as the event-study evidence of [loughran2011](#) would predict, but neither sign nor significance reverses. The split between an active LitDensity and a null UncDensity is preserved.

Table 4.4: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (365 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.6112*** (+14.37)	+0.5731*** (+12.60)	+0.5935*** (+13.41)	+0.5680*** (+12.34)	+0.5657*** (+12.25)
$\ln(\text{Total Assets})$	-0.0339*** (-4.76)	-0.0346*** (-4.94)	-0.0346*** (-4.91)	-0.0349*** (-5.00)	-0.0337*** (-4.76)
Leverage	-0.0116 (-0.81)	-0.0026 (-0.17)	-0.0139 (-0.94)	-0.0051 (-0.34)	-0.0056 (-0.37)
ROA	-0.4402*** (-7.45)	-0.3901*** (-7.20)	-0.4552*** (-7.54)	-0.4054*** (-7.43)	-0.4101*** (-7.42)
Asset Growth	+0.0728** (+2.49)	+0.0586** (+2.29)	+0.0684** (+2.38)	+0.0579** (+2.25)	+0.0568** (+2.20)
$\ln(\text{Words}^{1A})$		+0.0645*** (+5.15)		+0.0558*** (+4.53)	+0.0607*** (+4.93)
RiskDensity			-3.8105*** (-5.42)	-2.1913*** (-3.55)	
UncDensity					-0.8552 (-1.08)
LitDensity					-3.3905*** (-3.91)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	6,172	6,172	6,172	6,172	6,172
Firms	535	535	535	535	535
Adj. R^2	0.729	0.734	0.731	0.735	0.735

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 365 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its decomposition (H3) into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t -statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.4 Multi-horizon table

Table 4.5 reports the H1+H3 decomposition specification of equation 3.7 at six post-filing horizons: $h \in \{5, 10, 30, 90, 180, 365\}$ calendar days. Each column is a separate regression on the full panel; financial controls and fixed effects are unchanged.

Table 4.5: Post-Filing Volatility at Alternative Horizons

	(1) 5d	(2) 10d	(3) 30d	(4) 90d	(5) 180d	(6) 365d
$\ln(\sigma_{i,t})$	+0.1008*** (+3.62)	+0.1249*** (+4.55)	+0.1321* (+1.75)	+0.2122** (+2.24)	+0.3533*** (+4.19)	+0.5657*** (+12.25)
$\ln(\text{Words}^{1A})$	+0.1953*** (+8.78)	+0.1524*** (+6.72)	+0.1298*** (+7.16)	+0.1155*** (+5.93)	+0.0962*** (+5.11)	+0.0607*** (+4.93)
UncDensity	+0.9530 (+0.27)	-2.1555 (-1.04)	-1.2779 (-0.62)	-0.7636 (-0.45)	-0.6946 (-0.55)	-0.8552 (-1.08)
LitDensity	-7.0957*** (-3.76)	-6.6115*** (-2.88)	-6.1591*** (-3.38)	-6.5495*** (-3.94)	-5.1636*** (-4.02)	-3.3905*** (-3.91)
Financial controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,690	5,648	5,691	5,691	5,691	6,172
Firms	535	535	535	535	535	535
Adj. R^2	0.156	0.354	0.507	0.575	0.643	0.735

Note: Each column re-estimates the H1 + H3-decomposition specification replacing the dependent variable with $\ln(\sigma_{i,t+1}^{[h]})$ measured over a post-filing window of length h . The lagged dependent variable is the same firm's log-volatility computed over the prior filing year's window of identical length. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A; together they sum to RiskDensity (H2). Financial controls (size, leverage, ROA, asset growth) are included but suppressed. Industry and year fixed effects are absorbed; t -statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Three patterns emerge. First, $\ln(\text{Words}^{1A})$ enters with a positive and highly significant coefficient at every horizon, ranging from +0.195 at 5 days to +0.061 at 365 days, with t -statistics between +4.93 and +8.78. The volume association is robust across horizons. Second, LitDensity enters with a negative coefficient at every horizon and is significant at the 1% level from 10 days onward; the t -statistics range from -2.88 at 10 days to -4.02 at 180 days. Third, UncDensity is small in absolute value and statistically indistinguishable from zero at every horizon, with $|t| \leq 1.1$ at all windows. The qualitative pattern of Column (5) of Table 4.3 therefore replicates at every horizon the data admit: the negative association between risk-word language and post-filing volatility is a litigious-language association, not an uncertainty-language one.

The lagged-dependent-variable coefficient also rises with horizon, from about 0.030 at 5 days

to 0.566 at 365 days, reflecting the mechanical fact that longer windows for both the dependent and the lagged variable share more of their underlying daily-return sample. This is a feature of the data, not of the specification.

4.5 The boilerplate decomposition

The headline result of Column (5) of Table 4.3 and the consistent across-horizon pattern of Table 4.5 together support a sharp interpretation of what the H2 “specificity effect” actually is. The composite RiskDensity used in much of the prior literature (**loughran2011**; **kravet2013**) pools two LM word lists with very different production technologies. Our decomposition shows that the negative association of RiskDensity with post-filing volatility is driven entirely by the LM-Litigious sub-list. Once the litigious component is partialled out, the LM-Uncertainty component has no incremental association with volatility at any horizon.

The magnitude is economically meaningful. A one-standard-deviation rise in LitDensity (≈ 0.005) corresponds to a $0.005 \times 6.16 \approx 2.8\%$ decrease in 30-day annualised volatility. The interquartile range of LitDensity (≈ 0.006) implies an $\approx 3.8\%$ swing in volatility from the lowest to the highest litigious-density quartile.

The interpretation aligns with the boilerplate-disclosure literature reviewed in Section 1.4. **nelson2007** and **cazier2021** document that litigious-list language in mandatory filings is dominated by standardised cautionary boilerplate inserted by legal counsel under the safe-harbour regime of the 1995 Private Securities Litigation Reform Act, while **loughran2015** stress that uncertainty and litigious wordings are conceptually distinct constructs. Firms whose Item 1A is heavier in litigious wording are therefore plausibly firms with more institutionalised compliance functions and more legally supervised disclosure processes, characteristics that prior work (**nelson2007**; **cazier2021**; **beatty2019**) associates with lower realised risk. The Item 1A length effect (H1) and the litigious-density effect (H3) are then two distinct firm-level signals: the former captures how much management has chosen to disclose, the latter captures the legal-procedural style in which it is disclosed.

A caveat on identification is reserved for Section 5.3 of the next chapter. **H3 is supported in the cross-section**, with the LitDensity sign holding at every horizon and on both COVID sub-samples; whether it survives within-firm absorption is examined separately in the robustness chapter.

4.6 Economic magnitude

Table 4.3 translates straightforwardly into economic magnitudes because the dependent variable is in logs and the regressors are either in logs or are bounded densities.

Length. A one-standard-deviation rise in $\ln(\text{Words}^{1A})$ (≈ 0.64 , corresponding to a $\approx 90\%$ increase in word count) implies an $0.64 \times 0.130 \approx 8.3\%$ increase in 30-day annualised volatility (using the Column 5 estimate). Going from the 25th to the 75th percentile of Item 1A length implies an $0.78 \times 0.130 \approx 10.1\%$ increase in volatility. Relative to a sample-mean annualised volatility of ≈ 0.26 , this is roughly 2.6 percentage points of annualised volatility—an economically substantive effect well within the range of disclosure effects reported in [loughran2011](#); [campbell2014](#)<empty citation>.

Composite specificity (H2). A one-standard-deviation rise in RiskDensity (≈ 0.006) implies a $0.006 \times (-3.88) \approx -2.3\%$ change in 30-day annualised volatility. The interquartile range of RiskDensity is ≈ 0.008 , implying an $\approx -3.1\%$ swing in volatility from the lowest to the highest density quartile.

Litigious sub-component (H3). A one-standard-deviation rise in LitDensity (≈ 0.005) implies a $0.005 \times (-6.16) \approx -2.8\%$ change in 30-day annualised volatility—of the same order of magnitude as the composite H2 effect, despite LitDensity being only one-third of RiskDensity by share. The fact that the entire composite effect can be relocated onto a sub-list one-third its size is the empirical content of the H3 decomposition: the negative “specificity” effect of generic risk vocabulary is in fact concentrated in cautionary legal-process boilerplate.

Why the dictionary-only ratio understates the specificity channel. The univariate Column (3) coefficient on RiskDensity is -7.78 , twice the Column (4) joint estimate of -3.88 . The reason is the negative correlation between the two text variables (Section 4.1): longer Item 1A sections tend to be slightly less dense, so omitting length from the regression loads part of the (positive) volume effect onto a variable (RiskDensity) that is itself negatively correlated with length, exaggerating the negative coefficient. The joint specification of equation 3.6 disentangles the two channels and the H3 decomposition (equation 3.7) localises the resulting effect inside the LM lexicon.

5. Robustness

This chapter checks that the headline results of Chapter 4 survive plausible variations in the specification. Section 5.1 reviews the multi-horizon evidence already documented in Section 4.4 from a robustness angle. Section 5.2 re-estimates the H1+H3 decomposition specification on the pre-COVID and COVID-era sub-samples. Section 5.3 replaces industry fixed effects with firm fixed effects, which absorbs all time-invariant firm-level heterogeneity and isolates the within-firm year-to-year variation in the three text variables. Section 5.4 closes the chapter with a transparent account of the seven alternative H3 candidates that were tested and rejected before settling on the LM-list decomposition.

5.1 Alternative post-filing windows

The headline specification measures post-filing volatility over a 30-day window. One month is short enough to capture the bulk of the post-filing reaction documented by **loughran2011** but long enough that the volatility estimate is built on roughly 21 trading days of returns. To check the choice of window we re-estimate the H1+H3 specification of equation 3.7 at five additional horizons (5, 10, 90, 180, and 365 days) and inspect the resulting series of coefficients. Table 4.5 reports all six columns side by side; the discussion in Section 4.4 is repeated here from a robustness angle.

The pattern is consistent in three respects. First, all three text coefficients keep the predicted sign at every horizon—positive for $\ln(\text{Words}^{1A})$, negative for LitDensity, and statistically indistinguishable from zero for UncDensity—so neither H1 nor H3 reverses for any plausible choice of window. Second, the LitDensity coefficient is significant at the 1% level from 10 days onward, with $|t|$ between 2.88 (10 days) and 4.02 (180 days). Third, the UncDensity coefficient never approaches statistical significance, with $|t| \leq 1.1$ at every window. The 30-day choice for the headline specification is therefore not a window in which the results happen to be sharpest—they are sharp at every horizon the data admit.

Both coefficient series also decline approximately monotonically in h , which is the empirical signature of an event-driven signal that is incorporated into prices over weeks rather than years (**loughran2011**; **tetlock2007**). The thesis does not fit a formal exponential decay because the coefficient series is short and the across-horizon error terms are correlated by the sharing of underlying daily returns, which would understate the bootstrap CI on any half-life estimate. The qualitative attenuation is nonetheless visible by inspection of Table 4.5.

5.2 Pre-COVID and COVID-era sub-samples

The 2020 fiscal year and its aftermath were a structural shock for both equity volatility and corporate disclosure: average annualised volatility roughly doubled in the spring of 2020 and Item 1A sections grew systematically longer as firms added pandemic and supply-chain language. To check that the headline results are not driven by either the pre-COVID baseline or the COVID-era spike, we re-estimate the H1+H3 decomposition specification (equation 3.7) separately on the two sub-periods. Filings with fiscal year ≤ 2019 form the pre-COVID sample (3,243 firm-years) and filings with fiscal year ≥ 2020 form the COVID-era sample (2,448 firm-years). All other features of the specification are unchanged. Table 5.1 reports the result.

Table 5.1: Sub-Period Robustness (30-day horizon)

	(1) Pre-COVID (2010–2019)	(2) COVID era (2020–2024)
$\ln(\sigma_{i,t})$	+0.1271 (+1.21)	+0.1313* (+1.72)
$\ln(\text{Words}^{1A})$	+0.1025*** (+6.06)	+0.1715*** (+8.92)
UncDensity	−2.8830 (−1.17)	+0.8033 (+0.33)
LitDensity	−5.4863** (−2.55)	−9.8819*** (−4.85)
Financial controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	3,243	2,448
Firms	470	527
Adj. R^2	0.529	0.426

Note: H1 + H3-decomposition specification re-estimated on two non-overlapping sub-samples. The dependent variable is $\ln(\sigma_{i,t+1}^{[30]})$. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A. All financial controls and SIC2+year fixed effects are included. Standard errors are two-way clustered by firm and fiscal year. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The two active text channels are present in both sub-periods with the predicted signs and at the same order of magnitude. Length is significant at the 1% level in both ($\beta = +0.103$, $t = +6.06$ pre-COVID; $\beta = +0.172$, $t = +8.92$ COVID era). LitDensity is negative and significant in both ($\beta = -5.49$, $t = -2.55$ pre-COVID; $\beta = -9.88$, $t = -4.85$ COVID era). UncDensity is statistically indistinguishable from zero in both samples, consistent with the headline finding that uncertainty wording adds no explanatory power once length and litigious wording are controlled for. The two active coefficients (length and LitDensity) are larger in absolute value in the COVID-era sample, consistent with the higher overall level of

volatility in that period giving more scale to the disclosure signal, but the qualitative pattern is identical. The headline finding is not a COVID artefact.

5.3 Within-firm identification: firm fixed effects

The headline specification absorbs two-digit SIC industry and fiscal-year fixed effects. Coefficients are therefore identified from variation *within an industry-year cell*, which removes the most plausible confounders (industry-level differences in risk vocabulary and macroeconomic time effects) but still allows permanent firm-level traits to drive the result. To rule that out we replace the industry fixed effects with firm fixed effects, so that identification comes only from *within-firm* year-to-year variation in the three text variables. Table 5.2 reports the result at the 30-day horizon.

Table 5.2: Within-Firm Identification (Firm + Year FE, 30-day horizon)

	Estimate
$\ln(\sigma_{i,t})$	−0.1059*** (−6.77)
$\ln(\text{Words}^{1A})$	+0.0835*** (+3.31)
UncDensity	−1.4559 (−0.55)
LitDensity	+0.9561 (+0.35)
Financial controls	Yes
Firm FE	Yes
Year FE	Yes
Observations	5,691
Firms	535
Adj. R^2	0.583

Note: H1 + H3-decomposition specification with firm and fiscal-year fixed effects absorbed. Identification of $\ln(\text{Words}^{1A})$, UncDensity and LitDensity comes only from within-firm variation across years. Standard errors clustered by firm. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The volume channel survives the firmer specification: $\hat{\beta}^{\text{len}} = +0.084$ ($t = +3.31$, $p < 0.001$). A within-firm one-standard-deviation rise in $\ln(\text{Words}^{1A})$ still predicts a meaningful rise in post-filing volatility, so the H1 result is not a cross-sectional artefact—firms that grow their Item 1A from year to year do experience higher subsequent volatility.

Both LM sub-densities, by contrast, are absorbed by firm fixed effects: $\hat{\beta}^{\text{unc}} = -1.46$ ($t = -0.55$) and $\hat{\beta}^{\text{lit}} = +0.96$ ($t = +0.35$). The interpretation is straightforward and economically informative. The composition of an Item 1A’s risk wording—both the uncertainty share and

the litigious share—has very little within-firm variation across years, because the section is largely recycled from one filing to the next (**cazier2021**) and the legal advisers who shape its cautionary language change slowly. A specification that throws away the cross-sectional variation has therefore almost no remaining variance in either density to identify a coefficient from.

This pattern is consistent with the production-side reading of litigious language given by **nelson2007**; **cazier2021**; **beatty2019**<empty citation>: cautionary boilerplate is a quasi-permanent firm-level disclosure-style trait that proxies for an institutionalised, legally supervised compliance function. It is precisely such a permanent firm trait that would be cross-sectionally informative but within-firm null.

A single thesis cannot adjudicate between the two readings of LitDensity that the data are consistent with: (i) it captures a genuine causal between-firm effect of legal-style boilerplate on perceived risk, or (ii) it correlates with an unobserved firm-level trait (e.g. governance quality, legal-counsel seniority, board composition) that itself drives volatility. The firm-FE result rules out the second reading only as a within-firm driver. The thesis reports both findings transparently and reads the headline result as a between-firm regularity that aligns with the **nelson2007**; **cazier2021**; **beatty2019**<empty citation> interpretation, without claiming a within-firm causal channel.

Bottom line. Across alternative horizons and sub-periods, the qualitative picture of Chapter 4 is preserved: longer risk-factor sections predict higher post-filing volatility, and—wherever cross-sectional variation in disclosure style is available—denser litigious-list sections predict lower post-filing volatility, while denser uncertainty-list sections do not. H1 clears all robustness checks including firm-FE; H2 and H3 clear the cross-sectional and sub-period checks but are absorbed by firm-FE because both the composite RiskDensity and its two LM sub-components are quasi-permanent firm-level traits.

5.4 What did not work: alternative H3 candidates

In the spirit of full disclosure, this section documents the seven text-based H3 candidates that were estimated on the same panel and rejected before the LM-list decomposition was adopted. All seven specifications add a third Item 1A or MD&A text variable to the H1+H2 joint specification of equation 3.6 and ask whether it adds incremental explanatory power for $\ln \sigma_{i,t+1}^{[30]}$.

Year-on-year textual similarity (Item 1A). A TF-IDF cosine similarity between consecutive years' Item 1A sections, in the spirit of **brown2011**<empty citation>'s MD&A modification score, was estimated as a measure of disclosure “staleness”. The coefficient was

not statistically distinguishable from zero in the joint specification, with the predicted sign nonetheless reversing across horizons.

MD&A LM-Negativity. The share of LM-Negative words in the Management’s Discussion and Analysis section was estimated as a functionally distinct sentiment proxy. It did not add incremental explanatory power once Item 1A length was controlled for.

Year-on-year change in Item 1A length (ΔWords). The first difference of $\ln(\text{Words}^{1A})$, motivated by the [lyle2023<empty citation>](#) “changes” framing, was statistically indistinguishable from zero in the joint specification.

FinBERT-tone (four functional forms). A pretrained FinBERT-tone classifier ([yiyanghkust/finbert-tone](#)) was run on every Item 1A in the panel. Four candidate signals were constructed from the per-sentence tone probabilities and aggregated to the section level: (i) $P(\text{negative})$, (ii) the signed difference $P(\text{positive}) - P(\text{negative})$, (iii) the share of neutral sentences $P(\text{neutral})$, and (iv) the “intensity” measure $1 - P(\text{neutral})$. None of the four adds statistically significant explanatory power to the joint specification at the 30-day horizon, with $|t| < 1$ in all four cases. The detailed null results are not tabulated in the interest of brevity, but the FinBERT scoring pipeline (`scripts/finbert_score_mac.py`) and its per-firm output are preserved in the project repository for future replication.

The negative results have an intuitive reading. The TF-IDF and FinBERT signals capture firm-specific narrative changes whose informational content is, on this panel, dominated by the much stronger length signal of H1. The LM-list decomposition succeeds where these alternatives fail because it identifies a structural composition of the H2 specificity effect rather than an additional firm-specific signal that competes with H1.

Conclusion

Summary of findings

On an unbalanced panel of 535 large U.S. non-financial firms over fiscal years 2010–2024, the joint and decomposition specifications of Chapter 3 deliver three findings. **H1 is supported:** holding firm fundamentals, industry, and year fixed, the natural log of the Item 1A word count is positively associated with post-filing log-volatility at the 30-day horizon ($\hat{\beta} = +0.130$, $t = +7.70$ in the H1+H3 specification). The volume effect of **campbell2014**<empty citation> is therefore present in a panel that is different in size and time period from the original, and survives within-firm absorption. **H2 is supported:** holding length and the same controls fixed, the composite LM risk-word density of Item 1A is negatively associated with post-filing log-volatility ($\hat{\beta} = -3.88$, $t = -2.74$). The specificity effect of **hope2016**<empty citation>, translated into dictionary terms, is present once the volume channel is partialled out. **H3 is supported:** decomposing the composite RiskDensity into its two LM sub-components reveals that the entire negative effect is carried by the LM-Litigious sub-list ($\hat{\beta}^{\text{lit}} = -6.16$, $t = -3.35$), while the LM-Uncertainty sub-list is statistically indistinguishable from zero. The pattern survives at every post-filing horizon between 5 and 365 days and on both COVID-era and pre-COVID sub-samples. The interpretation aligns with the **nelson2007**; **cazier2021**; **beatty2019**<empty citation> reading of litigious-list language as standardised cautionary boilerplate characteristic of mature, legally supervised disclosure environments.

Methodological contribution

The thesis’s methodological contribution has two layers. First, the dictionary-based risk-word density used in much prior work mixes two opposite-signed channels into one statistic through a shared denominator. The univariate density coefficient at the 30-day horizon (-7.78) is roughly twice the joint estimate (-3.88); the gap is exactly the part of the volume channel that loads onto density through the negative cross-sectional correlation between length and density. Entering the two regressors jointly in a single specification recovers two effects that the dictionary-only ratio hides.

Second, the H2 “specificity” effect itself pools two LM word lists with very different production technologies. The LM-list decomposition (equation 3.7) shows that the negative coefficient on the composite is driven entirely by the LM-Litigious sub-list, while the LM-Uncertainty sub-list adds no explanatory power. To our knowledge no prior panel study of post-filing realised volatility estimates the LM-Litigious density of Item 1A as a stand-alone regressor; the closest references (**loughran2011**; **kravet2013**; **campbell2014**; **lyle2023**) either aggregate the two sub-lists, hand-code their own risk categories, or focus on year-on-year changes. The

decomposition therefore localises a known specificity effect to a specific LM sub-list and gives it a concrete interpretation in light of the boilerplate-disclosure literature (**nelson2007; cazier2021; beatty2019**).

Limitations

Four caveats apply. First, the LM dictionary is a bag-of-words measure: it does not capture irony, negation, or context, so the *specificity* construct used here is a coarse proxy for the named-entity-based measure of **hope2016<empty citation>**. Second, the identification is correlational. The within-firm robustness check (Section 5.3) preserves the volume effect but absorbs both LM sub-densities, which is consistent with risk-vocabulary composition being a quasi-permanent firm-style trait (**cazier2021**) but means H2 and H3 cannot be claimed to hold in the time series of a given firm. Third, the sample is restricted to large U.S. non-financial firms with continuous XBRL coverage; results may not generalise to small caps, financials, or non-U.S. filers. Fourth, the boilerplate-disclosure interpretation of LitDensity follows the **nelson2007; cazier2021; beatty2019<empty citation>** literature but is not directly tested with a measure of safe-harbour text recycling; it is consistent with the data rather than demonstrated by them.

Suggested extensions

Three extensions follow naturally. First, the named-entity pipeline of **hope2016<empty citation>** would replace the dictionary-based specificity proxy with a direct measure of risk-factor specificity and would provide a stronger test of H2. Second, a forward-looking statement classifier in the spirit of **li2010<empty citation>**, possibly built on a fine-tuned FinBERT-FLS encoder, would let the H3 literature’s distinction between forward-looking cautionary language and backward-looking litigious narrative be tested separately; this would also extend the seven null FinBERT-tone results documented in Section 5.4 to the functional form (FLS classification) that the FinBERT family is actually trained for. Third, an explicit text-recycling measure (year-on-year overlap of LM-Litigious sentences) would test the boilerplate-disclosure interpretation of the H3 result directly, rather than inferring it from the cross-sectional pattern of LitDensity coefficients.

Appendices

A. Variable Definitions and Data Pipeline

This appendix collects supplementary tables referenced in the main text. Section A.1 presents the diagnostics behind the choice of annual filing frequency (Section 2.2). Section A.2 reports the per-firm XBRL tag assignments from the tag-harmonisation procedure (Section 2.3).

A.1 Filing Frequency Diagnostics

The following tables supplement the discussion in Section 2.2.

Table A.1: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.2: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	~52% substantive
Risk Factors content	Median ~30,000 words	Median ~9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	~11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
Item 1A length (H1)	Feasible (annual word count)	Limited (52% substantive)
Item 1A risk-word density (H2)	Feasible (annual LM ratio)	Limited (52% substantive)
Literature alignment	Standard	Uncommon
Expected observations	~6,400 firm-years	~19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required by SEC Regulation S-K Item 1A only when there are “material changes” from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.1, 2.1, and 2.2 for detailed results.

Table A.3: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: “MD&A Found” and “Risk Subst.” count filings with >200 substantive words. “Risk Boilerplate” counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.2 XBRL Tag Provenance

Table A.4 reports, for each metric with multiple candidate XBRL tags, the number and share of firms locked to each tag by the per-firm tag-locking procedure described in Section 2.3.

Table A.4: Per-firm XBRL tag distribution

Metric	Locked XBRL tag	Firms	%
Net income	NetIncomeLoss	502	91.6
	ProfitLoss	45	8.2
	IncomeLossFromContinuingOperations	1	0.2
Operating cash flow	NetCashProvidedByUsedInOperatingActivities	530	96.7
	...OperatingActivitiesContinuingOperations	18	3.3
Stockholders' equity	StockholdersEquityIncl...NoncontrollingInterest	318	58.6
	StockholdersEquity	225	41.4

Notes. For each (firm, metric) pair, the tag with the highest within-firm year-count is selected; values from other tags are set to missing. Firm counts are based on up to 548 firms in the tag-locking stage; the stockholders' equity row sums to 543 because five firms report no equity tag at all. See Section 2.3 for the rationale.

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction Validation (Section 2.4.5)

Tool. Google Gemini 2.5 Flash (accessed via the OpenAI-compatible Gemini API endpoint), with temperature set to 0 for deterministic output.

Purpose. An independent, automated quality audit of the SEC 10-K section extraction pipeline. The model evaluated 200 extracted sections (100 Item 1A, 100 Item 7) against a structured five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) on a three-level scale (Pass / Minor / Fail).

Contribution and verification. The evaluation rubric, sampling strategy, prompt design, and statistical analysis (Wilson confidence intervals) were all designed by the author. The model's only role was applying the rubric to each extracted section. All Fail verdicts were manually spot-checked against the original SEC filings and confirmed to be correct. Results are in Table 2.3.

Code Development

Tool. AI-powered code completion (integrated in VS Code).

Purpose. Code completion during the development of the data-pipeline scripts that build the panel and extract the 10-K text sections.

Contribution and verification. All generated code was reviewed, tested, and modified by the author. The overall architecture, variable definitions, and analytical choices are the author's own.

Writing Assistance

Tool. AI-powered writing assistant (integrated in VS Code).

Purpose. Grammar and style suggestions for English-language prose in the thesis text.

Contribution and verification. All substantive content, argumentation, and interpretation are the author's own. The tool was used only for language polishing; every suggestion was reviewed before acceptance.